Bud-JET Airlines: Cloud Data Infrastructure Design

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1. Mission

Bud-JET Airlines aims to design and implement a **scalable**, **resilient**, and **secure** cloud infrastructure. This infrastructure will integrate various data sources to support airline operations, enhance internal workflows, and maintain high performance and reliability.

2. Objectives

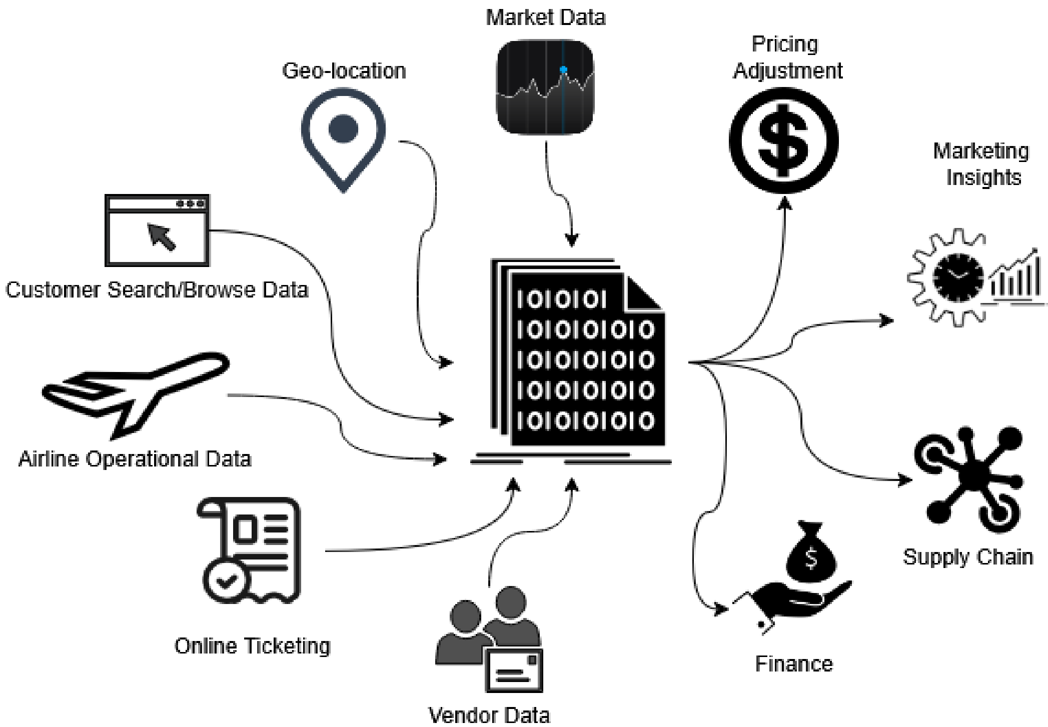
To meet the overarching mission, Bud-JET Airlines sets out the following strategic objectives:

- **Scalable Infrastructure**
  Design a system capable of adapting to demand fluctuations while maintaining service availability during peak usage.
- **Data Consistency**
  Ensure data integrity and reliability across systems, supporting vital airline operations like booking, scheduling, and inventory.
- **Centralized Repository**
  Create a single source of truth for operational data, enabling efficient reporting and real-time analytics.
- **System Monitoring**
  Incorporate tools for monitoring, logging, and alerts to ensure system health and allow swift issue resolution.

3. Design Vision

The envisioned cloud ecosystem will be **future-proof**, supporting Bud-JET Airlines with:

- Integration of diverse data sources (both streaming and batched)
- Scalable analytics and reporting capabilities
- Real-time insights to drive both operational efficiency and marketing strategies.



4. Data Sources

4.1 Batched Sources

Source	Nature	Format
Online Ticketing	Structured, Real-time	CSV, JSON
Vendor Data (OTA)	Structured, Batched/RT	CSV, JSON
Airline Operational DB	Structured, Batched	CSV, JSON
Historical Sales & Bookings	Structured, Batched	CSV, JSON
Competitor & Market Data	Semi-Structured, Mixed	CSV, JSON

4.2 Streaming Sources

Source	Nature	Format
Vendor Data (OTA)	Structured, Mixed	CSV, JSON
Customer Search & Browsing	Unstructured, Streaming	XML, TXT, Logs

5. Data Sinks

Processed data is utilized to drive actionable insights across departments:

- **Operational Efficiency Reports** – Supply Chain
- **Customer Behavior Insights** – Marketing
- **Demand Forecasting** – Supply Chain & Pricing
- **Pricing Optimization** – Finance
- **Route Optimization** – Supply Chain
- **Competitor Market Analysis** – Finance & Marketing

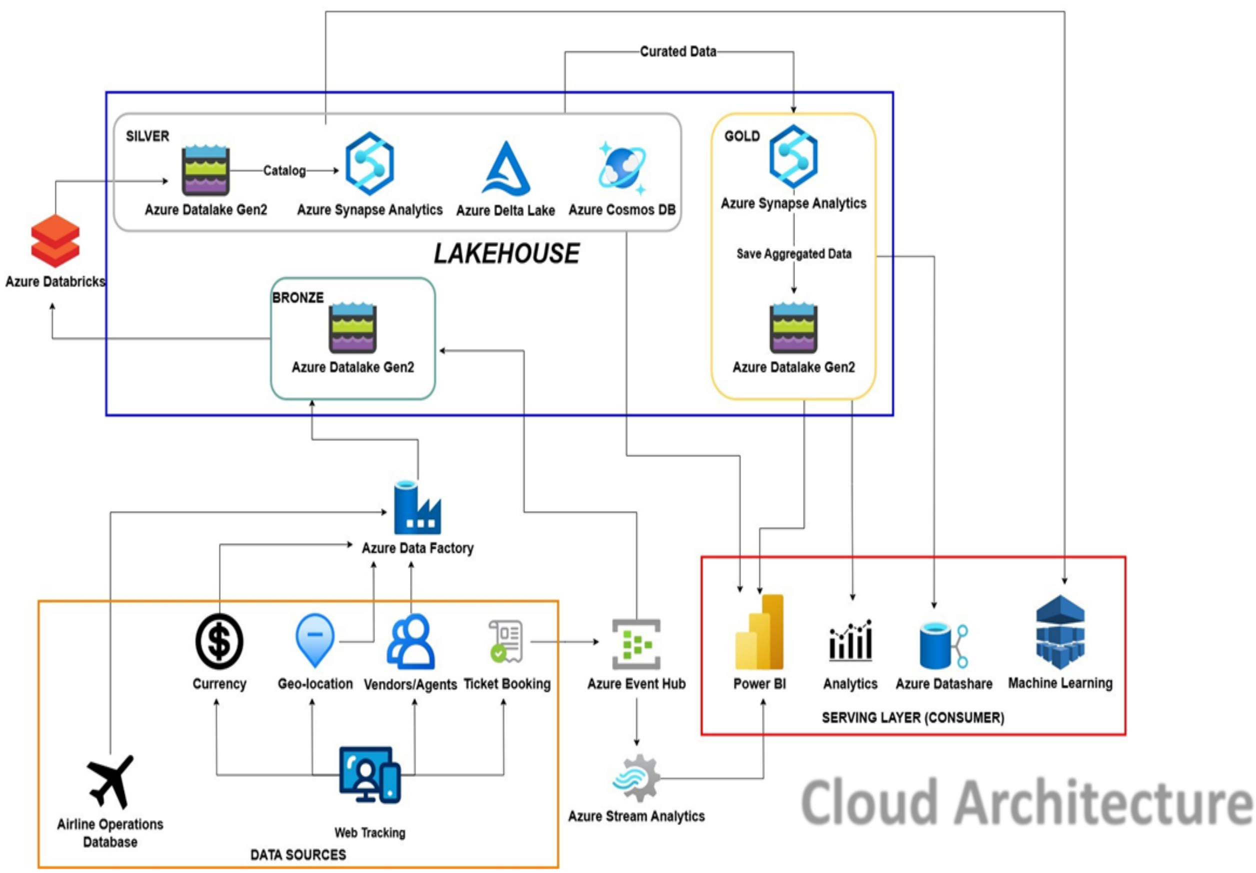
6. Updated Data Sources

Source	Nature	Format
Operational DB	Structured, Batched	CSV, JSON
Web Tracking	Structured, Batched	CSV, JSON
Currency	Structured, Batched	CSV, JSON
Geo-location	Semi-Structured, Batched	CSV, JSON
Vendors/Agents	Structured, Batched	CSV, JSON
Ticket Booking	Structured, Streaming/RT	CSV, JSON

7. Serving Layer

Bud-JET's serving layer is the interface for data access, insights, and distribution:

- **Power BI** – Business Intelligence dashboards
- **Azure Data Share** – Secure data distribution
- **Machine Learning Pipelines** – Predictive analytics and automation
- **Analytics Layer** – Actionable and real-time insights



8. Data Processing Layers

8.1 Silver Layer – Data Curation

Activities:

- Data inspection
- Data cleaning
- Data verification

Flow:

1. Define KPIs
2. Shortlist Observations
3. Summarize Data
4. Visualize Insights

8.2 Gold Layer – Data Aggregation

- Aggregation performed on an hourly/daily basis
- Data structured for final dashboards and executive analytics

9. Pipeline Failure Strategy

To maintain robustness and minimize data loss, the infrastructure includes advanced retry and recovery mechanisms.

Retry & Resilience Features:

- Maximum Retry Limits
- Backoff Intervals
- Timeout Handling
- Monitoring & Alerts
- Smart Re-run Capabilities

Recovery Plans by Pipeline Type:

Pipeline Type	Strategy Description
Batch Ingestion	Timestamp-based checkpointing; resume from last success
Streaming Ingestion	Session window tracking; resume from last known offset
Curation Pipeline	Resume from last successful curation; avoid duplication
Aggregation Pipeline	Recover from failed window using time-based strategy.

10. Conclusion

Bud-JET Airlines' cloud infrastructure is the cornerstone of its data-driven transformation. By incorporating:

- Strategic data pipelines
- Robust failure handling
- Scalable, secure architecture

The organization positions itself to deliver fast, reliable insights while staying agile in a competitive market. This modern infrastructure provides a solid foundation for innovation, enhanced customer experiences, and sustainable growth.